

ATHARVA AMBADE

San Jose, CA | 510-695-3744 | ambade.atharva@outlook.com | ambadeatharva.github.io | linkedin/ambadeatharva

SKILLS

Languages: Python, Java, C++, Go, SQL, TypeScript, JavaScript, Bash/Shell
Backend & Full Stack: Distributed Systems, RESTful APIs, Microservices, Spring Boot, FastAPI, Flask, React
AI Engineering: AI Agents, LLMs, RAG, PyTorch, LangChain, FAISS, Natural Language Processing
Cloud, DevOps & Infra: AWS, Kubernetes, Docker, Jenkins, Terraform, CI/CD, Observability (Splunk), Redis

EDUCATION

California State University, East Bay | Master's Degree in Computer Science May 2026
GPA: 3.8/4.0 | Coursework: Distributed Systems, Advanced AI, Machine Learning, Web Architecture

Savitribai Phule Pune University | Bachelor's Degree in Computer Engineering May 2021
GPA: 3.5/4.0 | Coursework: Data Structures, Algorithms, Computer Networks, Operating Systems

EXPERIENCE

AI/ML Research Assistant Hayward, CA
California State University, East Bay January 2025 - May 2026

- Implemented Binary Neural Networks and stochastic threshold techniques in **PyTorch** to resolve non differentiable gradient bottlenecks. Reduced memory footprint by **85%** while maintaining **94% validation accuracy**.
- Developed anomaly detection algorithms using **PyTorch** to identify fake student applications. Analyzed behavioral patterns to successfully flag over **2,000 ghost student profiles**.

Software Engineer Hayward, CA
California State University, East Bay | Student Assistant September 2024 - May 2026

- Built the student facing Reduced Course Load interface on **Salesforce** Education Cloud and migrated legacy administrative pages to **React**. Reduced manual processing time by **60%** for **10,000+ students**.
- Configured secure **OAuth 2.0** identity integrations across university portals to establish Single Sign On capabilities. Streamlined authentication workflows and improved responsive interaction across the ecosystem.
- Integrated an internal campus **AI chatbot** to parse documentation and route support tickets. Validated response accuracy and developed **Python** automation scripts to reduce administrative data entry by **40%**.

Software Engineer Pune, India
Accenture | Packaged App Development Analyst January 2022 - June 2024

- Designed and implemented **Python** batch ETL jobs for overnight financial reconciliation. Processed **15+ TB of data** per run and reduced job failure rates by **40%**.
- Built secure RESTful APIs using **Java Spring Boot** supporting a total system throughput of **10,000 Requests Per Second**. Resolved relational database query bottlenecks via B tree indexing to decrease **P99 latency by 35%**.
- Extended **Jenkins** and Terraform pipelines with zero downtime configurations to drop deployment lead times from **2 days to 4 hours**. Managed production instrumentation via custom **Splunk** observability dashboards.
- Built a **Python** and Scikit learn anomaly detection prototype for transaction fraud patterns. Presented the predictive model design at an internal innovation showcase to corporate banking stakeholders.

PROJECTS

Distributed MCP Server | Python, AsyncIO, Redis, Llama 3.2, LangChain | GitHub

- Engineered an asynchronous **Python** server utilizing **Llama 3.2** and LangChain to automate enterprise workflows with **95% natural language command accuracy** while reducing context token usage by **60%**.

Enterprise Document Extraction AI | Python, React, FastAPI, RAG, FAISS | GitHub

- Built a full stack **React** and FastAPI document ingestion application utilizing a **RAG** pipeline backed by FAISS vector storage to achieve **95% extraction accuracy** with a retrieval latency **under 200ms**.

Containerized Plagiarism Detector | Python, Kubernetes, Docker, MobileNetV2 | GitHub

- Evolved peer reviewed **ICINC 2021** machine learning research into a containerized microservice via **Docker**. Deployed the image processing model to **Kubernetes** utilizing HPA scaling to eliminate compute bottlenecks.